

Fault Diagnosis of Gearboxes Using Feature Extraction in Time Domain

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Abstract—In the vibration signals of a faulty gearbox, there exist not only the response excited by source of faults such as rotating frequencies of shafts, mesh frequencies of gear pairs, their higher harmonic components and modulated components, but also natural and random vibrations of system, components and parts, and noise interferences. Thus it is difficult to recognize and capture typical fault frequencies. For this purpose, an approach was proposed for fault diagnosis of gearboxes, by combining temporal feature extraction based on independent component analysis (ICA) and fault classification based on forward least squares approximation support vector machine (FLSA-SVM). Multiple ICA neural networks were batch trained by multi-classes vibration data separately, each corresponding to one of fault patterns of gearboxes. A feature extractor was structurally constructed by the trained ICA networks, followed by a post-processing unit of feature extraction based on one second-order measure, i.e. residual total correlation (RTC). Experimental results show that the proposed approach is efficient and effective for fault diagnosis of gearboxes, even if the used train sets are of small size. The integrated ICA-RTC feature extractor is able to extract low-dimensional representative features from time-domain data, and to overcome some difficulties in traditional frequency-domain feature extraction.

Index Terms—independent component analysis; forward least squares approximation support vector machine (FLSA-SVM); feature extraction in time domain; fault diagnosis of gearboxes.

I. INTRODUCTION

G^EARBOXES are widely used as the driving part in many rotating machines such as steam and wind turbines, generator and electromotor, etc. The major gearbox faults may include teeth surface damage (gear pitting, sapling and cracking), chipped or broken tooth, bearing-related faults and loose foundation, etc. [1-2]. Many researches have been made on fault diagnosis of gearboxes. Fault development or wear is usually accompanied by increased harmonics of the machine running frequency, increased amount of sidebands around structure-related fault frequencies and a shift in the amplitude distribution of the measured signals [3]. Local faults in a

gearbox are usually identified by detecting the presence of higher impulses in the vibration signal. Even if there is no fault, tooth contact may occur prematurely at the tip of the drive gear due to tooth deflection and the velocity mismatch between a pinion and a gear. This causes an impact at the time of contact which gives rise to the larger vibration at the gears meshing frequency [4]. It is also known that loose foundation often manifest itself as impulses in the time signal. Furthermore, gear-related faults, for example gear pitting or sapling, chipped or broken tooth and bearing-related faults, often manifest themselves as non-stationary modulation of the meshing frequencies by their higher-order harmonics and coupling frequencies under strong noise interference. Therefore, some techniques such as frequency-domain or time-frequency domain representation, time-domain averaging based denoising and/or filtering, feature level data fusion, etc. have been applied to gearbox fault diagnosis [5-7].

It should be noted that the current techniques for gearbox fault diagnosis usually require some special processing steps, for example frequency or time-frequency representation and filter, etc., to detect and capture some desired frequency bins (around the meshing frequencies, their higher-order harmonics and coupling frequencies) for feature extraction. However, these processes are often empirical and thus difficult to effectively and efficiently implement, suffered from some negative factors or effects such as structural resonance, tooth deflection, driving velocity mismatch, strong noise interference, nongaussianity vibration by impulse activity and information redundancy among multi-channel measurements [3]. These may lead to the current techniques unsuccessful in fault diagnosis of gearboxes.

In the present study, independent component analysis (ICA) is used as a preprocessing tool both for redundancy reduction and for data fusion. During batch training of ICA networks for feature extraction, every train sample is fused into multiple independent components, according to principle of redundancy reduction. Consequently, mixture structure embedded into the data in every train set is captured. These trained ICA networks are structurally united into a time-domain feature extractor with which feature vectors are further extracted, based on one simple second-order statistical measure: residual total correlation (RTC). The extracted feature vectors are then used for training and testing a support vector machine (SVM) classifier to implement final fault classification.

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II. FAULT DIAGNOSIS OF GEARBOXES USING FEATURE EXTRACTION IN TIME DOMAIN

A. The information-maximization approach for ICA

The basic problem tackled here is how to maximize the mutual information (MI) that the output $\mathbf{y}$ of a neural network processor contains about its input $\mathbf{x}$. Bell and Sejnowski [8] proposed a neural network based algorithm for ICA, Informax, in which they maximized the joint entropy, $H[g(\mathbf{W}\mathbf{x})]$, of the elements of the linear transform squashed by a sigmoid function, $g(\cdot)$. This sigmoid function is the cumulative density function (c.d.f) of the signal we are trying to extract. Even if the particular c.d.f is unknown, good results have been obtained for high kurtosis distributions by using the logistic function.

$$g(\mathbf{u}) = (1 + e^{-\mathbf{u}})^{-1}, \quad \mathbf{u} = \mathbf{W}\mathbf{x} + \mathbf{w}_0. \quad (1)$$

where $\mathbf{W}$ is the weight matrix for the ICA network, and $\mathbf{w}_0$ is a bias vector. For maximizing the entropy, weights are updated incrementally according to the gradient of the entropy. Thus, the resulting learning rules are

$$\Delta \mathbf{W} \propto [\mathbf{W}^T]^{-1} + (1 - 2\mathbf{y})\mathbf{x}^T, \quad (2)$$

$$\Delta \mathbf{w}_0 \propto 1 - 2\mathbf{y}. \quad (3)$$

B. Training of the ICA based networks for feature extraction

Given C train sets $\Gamma_c = \{\mathbf{x}^{(k)}, k = 1, 2, \dots, K\}$, $c = 1, 2, \dots, C$, where $\mathbf{x}^{(k)}$ is the k th train sample, c the number of train set belongs to different pattern class. L is the size of the train set. Using the Informax algorithm proposed by Bell and Sejnowski, we can obtain an estimate of linear transform $\mathbf{W}^{(k)}$ from the k th train sample $\mathbf{x}^{(k)}$, according to the following learning rule.

$$\Delta \mathbf{W}^{(k)} \propto \left[(\mathbf{W}^{(k-1)})^T \right]^{-1} + (1 - 2\mathbf{y}^{(k)}) (\mathbf{x}^{(k)})^T, \quad (4)$$

$$\Delta \mathbf{w}_0^{(k)} \propto 1 - 2\mathbf{y}^{(k)}. \quad (5)$$

where $\mathbf{W}^{(k-1)}$ is the estimated linear transform using the $(k-1)$ th train sample $\mathbf{x}^{(k-1)}$. $k = 2, 3, \dots, L$. Especially, $\mathbf{W}^{(0)}$ is an initial $M \times N$ linear transform, which is randomly selected and fixed before the training of ICA based networks. For every train sets Γ_c , the same initial linear transform is used, which may conform to either uniform or normal distribution. Although both of them are usable as an initial linear transform for the training of ICA based networks, it seems that better training effects are achieved by the later than that by the former, which can be observed in subsequent experiment for gear fault diagnosis.

Thus, we can obtain C trained ICA networks, $\mathbf{W}_c^{(K)}$, $c = 1, 2, \dots, C$, using the given train sets Γ_c . They will be structurally united into a multi-dimensional feature extractor for final time-domain feature extraction from other classes-labeled vibration samples.

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C. Low-dimensional quantitative feature extraction based on residual total correlation

The ICA based feature extraction consists of multiple trained ICA networks. The logic behind ICA based time-domain feature extraction is similar with the one by Govindan in [9] which applied ICA to electrogram classification during atrial fibrillation. When multidimensional vibration measurements belonging to one pattern class is passed through an ICA network which has been trained on measurements from that particular pattern class, the residual total correlation (RTC) between the output channels would be less than when it is passed through a network trained on a different class. We selected multi-channel vibration data from each pattern class to build up the train set which were used to batch train multiple ICA networks, each one being trained on one class of data. Then, they are structurally united into a multi-dimensional feature extractor for final feature extraction from other classes-labeled vibration samples. The RTC at the output of the ICA based feature extractor, when a vibration sample was presented at the input, formed the feature vector for that data, which is used to train a classifier, for example one neural network or SVM for performance of the final classification. The RTC at the output of one ICA network can be formulated as follows according to

$$V_{\text{RTC}} = \sqrt{\sum_{i=1}^N \sum_{j=1}^N \text{corrcoef}_{ij}^2}. \quad (6)$$

where corrcoef_{ij} is the correlation coefficient at zero lag between the i th output channel and the j th output channel. V_{RTC} is the RTC at the output of the ICA network. N is dimension of the presented train sample at the input of the network.

D. Forward least squares approximation support vector machine (FLSA-SVM)

Support Vector Machines (SVM) is a powerful methodology for solving problems in nonlinear classification, function estimation and density estimation which have also led to many other recent developments in kernel based learning methods in general. SVMs have been introduced within the context of statistical learning theory and structural risk minimization (SRM).

In a FLSA-SVM, the parameter m is interpreted as the number of nonzeros of Lagrangian multipliers, that is, the number of support vectors (SVs), which is seen more obviously in its Lagrangian by introducing the Lagrange

multipliers α_i ($i = 1, \dots, M$) for each of the equality constraints giving [10]

$$\Theta(\mathbf{w}, b, \alpha) = \frac{1}{2}(\mathbf{w}^T \mathbf{w} + b^2) - \sum_{i=1}^m \alpha_i [\mathbf{w}^T \phi(\mathbf{x}_i) + b - y_i] - \beta |\Gamma|. \quad (7)$$

The optimal point requires that

$$\mathbf{w} = \sum_{i=1}^m \alpha_i \phi(\mathbf{x}_i), \sum_{i=1}^m \alpha_i = b, \mathbf{w}^T \phi(\mathbf{x}_i) + b = y_i, |\Gamma| = m. \quad (8)$$

The linear equations can be further simplified to

$$(\mathbf{K} + \mathbf{1}) \mathbf{a} = \mathbf{D} \mathbf{a} = \mathbf{y}, \quad |\Gamma| = m. \quad (9)$$

where $\mathbf{K} \in \mathbf{R}^{M \times M}$ and $\mathbf{K}_{ij} = K(\mathbf{x}_i, \mathbf{x}_j)$, $\mathbf{y} = [y_1, \dots, y_M]^T$, $\alpha = [\alpha_1, \dots, \alpha_M]^T$, and $\mathbf{D} \in \mathbf{R}^{M \times M}$, where $\mathbf{D}_{ij} = \mathbf{K}_{ij} + 1$. It is worth attention that, in this paper, $\mathbf{D}$ in (9) is also referred to as $\mathbf{D}_m$ as it has m support vectors. Thus, if all $\mathbf{M}$ data samples are used as support vectors, then $\mathbf{D}$ becomes $\mathbf{D}_M$.

FLSA can be applied to train an FLSA-SVM using a kernel-based dictionary $\mathbf{D}_M = \{d_i = K(\cdot, j) + 1 \mid i = 1, \dots, M\}$ of candidate basis functions. Since m is defined to be the number of SVs, it thus achieves a more direct control of the sparseness of its solution [10].

III. EXPERIMENTAL RESULTS

Six-channel machine vibration used in this experiment was measured by a coupled electromotor-gearbox kit that is driven by an electromotor (1900RPM). The experimental set-up used for simulation of gearbox faults is shown in figure 1.

Measurements are repeated for the experimental setup. Four patterns, i.e. fault free, teeth surface damage, loose foundation and their combination (teeth surface damage plus loose foundation), are induced to the gearbox system separately. The data was measured under low pass filtered (analog) to 1 KHz and digitized at 20 kHz. We computed shaft rotating frequencies gear meshing frequencies, according to the given parameters of the used electromotor and gear combinations in our experimental system. The rotating frequencies (Hz) of driving, medium and driven shafts are 31.67, 7.04 and 1.36 respectively. The meshing frequencies (Hz) of two gear combinations (18/81 and 16/83) are 570 and 112.64, respectively. But, these frequency components can hardly be observed in the FFT spectrums except that of fault free and loose foundation, which have been masked by strong noise interference, especially when there exist gear-related faults. As the above analysis, some negative effects or factors, such as structural resonance, tooth deflection, driving velocity mismatch, impulse activity and information redundancy among multi-channel measurements, also contribute to such confusion on these eigen-frequencies.

The data from every class were divided into forty data segments, each segment of length 1024 samples. Thirty samples were randomly selected from all forty data segments in every pattern class to build up the train set. We batch trained four ICA networks, each corresponding to one class.

Initial iteration number was set as 1000 times for each single training step (using one train sample).

In the train processes of ICA networks, the RTC is used as a convergence measure to evaluate and optimize the training processes. Based on the trained ICA networks, we constructed two ICA based feature extractors (three-dimensional and four-dimensional) to implement time-domain feature extraction, from other twenty five samples. The RTC measure was used again. Here, it was just used as a measure for feature extraction, not for convergence evaluation. After feature extraction, we randomly showed one of feature samples from each feature set in Fig. 4. In Fig. 5, spatial distribution of all three-dimensional feature samples was also described, including three pattern classes, i.e. fault free (FF), teeth surface damage (TSD) and loose foundation (LF).

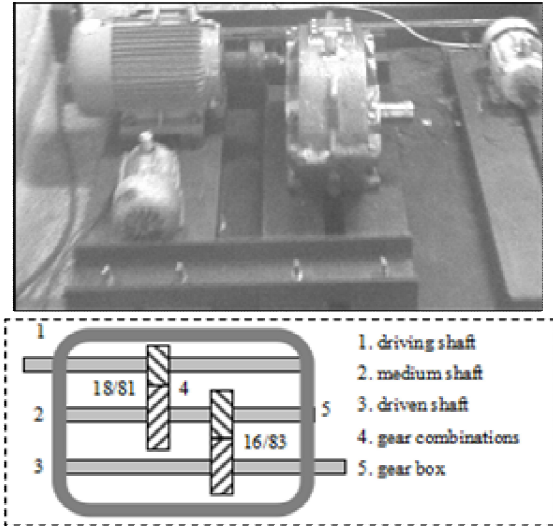


Fig. 1. Experimental setup and delaying gear-combinations

The accelerometer used for data collection is mounted as figure 2. The detailed positions are shown in Table I.

The time-domain waveforms of three patterns, i.e. FF, TSD and LF, were shown in figure 3. It is difficult to observe and capture distinct characteristics for further pattern classification, although they change at different modes.

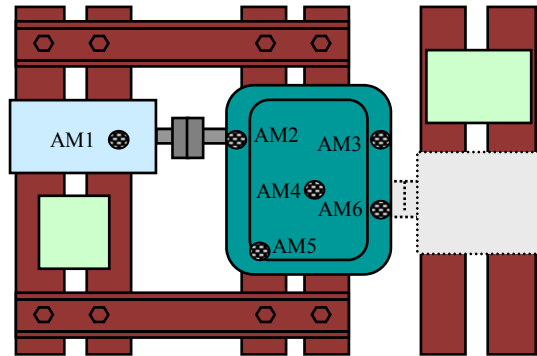


Fig. 2. Mounting alignment of measurement points

TABLE I
UNITS FOR MAGNETIC PROPERTIES

Symbol	Mounted position	Conversion from Gaussian and CGS EMU to SI ^a
AM 1	Outer machine case of the big electromotor	
AM 2	Gearbox driving shaft and left bearing	
AM 3	Gearbox Right bearing	
AM 4	Upper gearbox case	
AM 5	Left-bottom corner of gearbox case	
AM 6	Gearbox driven shaft and bearing	

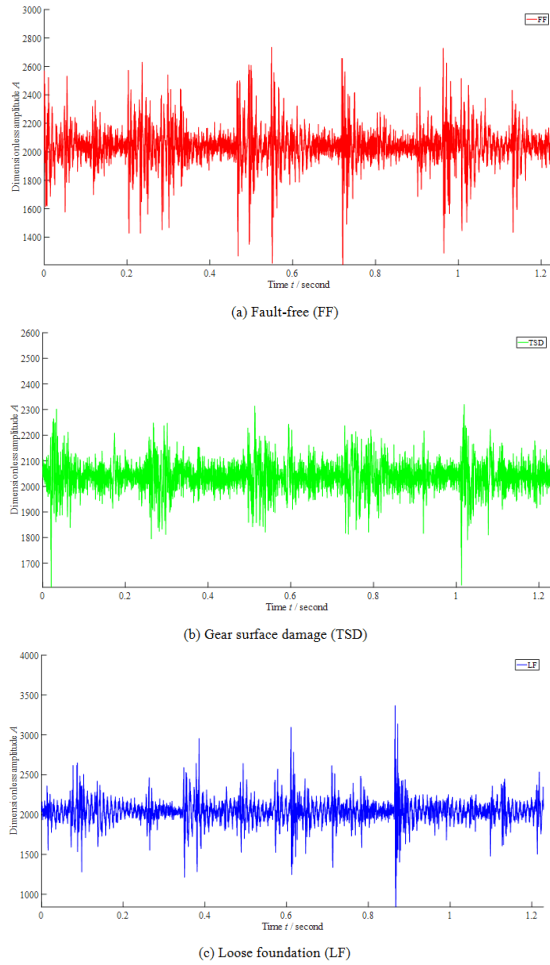


Fig. 3. Time-domain waveforms of three patterns

We used the FLISA-SVM software package by Matlab and C++ [11-12] to implement pattern classification with the extracted three-dimensional feature samples of different faults. The size of train set is 45, and the size of test set is 30, including three pattern classes, i.e. FF, TSD and LF, with 15 and 10 feature samples respectively. As comparison, we also used the LS-SVM toolbox by Matlab [13] to implement the same classification task. The classification results are shown in Table II together.

Using different coding schemes, the LS-SVM classifier shows different classification performances. The best classification result is achieved by the LS-SVM with the one versus all coding (OneVsAll) whose average accuracy reaches to 93.33%. Three other schemes are not good enough

for classification of the sparse multi-classes data. Their average accuracy is only 83.33%.

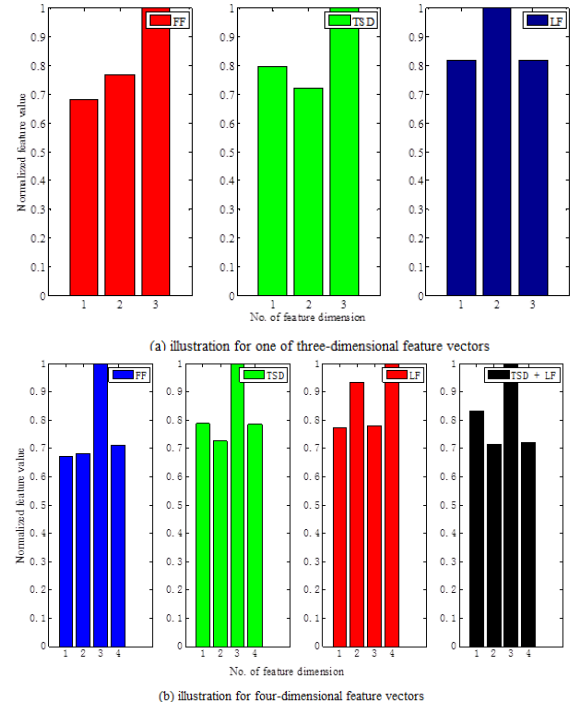


Fig. 4. Feature samples from different feature sets

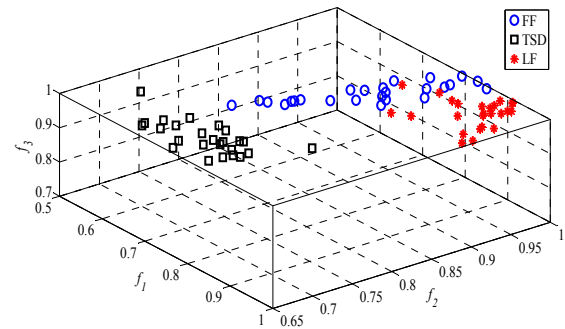


Fig. 5. Distribution of three-dimensional feature samples

TABLE II
THE CLASSIFICATION RESULTS

Classifier	Coding routines	Classes	Local accuracy	Average accuracy
LS-SVM	OneVsOne, MOC and ECOC ^a	FF	70%	83.33%
		FSD	100%	
		LF	80%	
	OneVsAll ^b	FF	90%	93.33%
		TSD	100%	
		LF	90%	
FLISA-SVM	AllVsAll ^c	All	/	96.67%

^a OneVsOne: One versus one coding; MOC: Minimum output coding; ECOC: Error correcting output coding;

^b OneVsAll: One versus all coding.

^c AllVsAll: All versus all coding

It can be also seen from Table II that the FLSA-SVM gives better classification. Its average accuracy is 96.67%. Especially, the FLSA-SVM runs faster (less than 0.8 second), because it uses the FLSA as the training algorithm, which avoids the term representing empirical risk, just that used in the LS-SVM. Such efficiency improvement will be more significant in case the size of data set is large.

IV. EXPERIMENTAL RESULTS

We propose an integrated method for automatic fault diagnosis of gearboxes, which combines ICA and FLSA-SVM. Experimental results show that both stable ICA based time-domain feature extractor and robust FLSA-SVM based fault classifier can be easily and reliably constructed even if the used data sets are of small size. The ICA based feature extraction in time domain makes a lot of empirical and thus difficult locations on some desired frequency bins (around the meshing frequencies, their higher-order harmonics and coupling frequencies) be avoided. Thus, the task of feature extraction becomes easy. The FLSA-SVM is also superior to the LS-SVM for classification of sparse multi-classes data. Under the premise of high classification accuracy, its computation cost is remarkably decreased. By the combinational use of ICA and FLSA-SVM, complex gearbox faults, including teeth surface damage, loose foundation or their combination (teeth surface damage plus loose foundation), can be accurately and efficiently classified. The study may be useful for future practical implementation of automatic fault diagnosis of gearboxes.

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